Dan Dean

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Work Experience

DevOps Engineer August 2024 — Present
Minitab *State College, PA*

- Provisioned and maintained critical infrastructure, including Azure Kubernetes Service (AKS) clusters, monolithic web servers, and SQL databases, supporting 5+ teams as a core platform backbone
- Assumed ownership of a custom Go-based Kubernetes Operator built with the Operator SDK, extending its CRD and reconciliation controller to dynamically provision and manage 100s of MySQL databases via the Azure Service Operator (ASO)
- Maintained and updated deployment infrastructure for 100s of live zero-downtime deployments using Helm, Azure DevOps Pipelines, and Terraform
- Spearheaded R&D for observability platforms, evaluating Azure Managed Grafana using Terraform and leading the production rollout of Datadog using the Datadog Operator and custom monitors, improving reliability across 3 regions
- Conducted cloud cost analysis to proactively identify anomalies, execute root-cause analysis, and implement remediation strategies to optimize resource efficiency

Learning Assistant August 2023 — May 2024
The Pennsylvania State University *State College, PA*

- Assisted students in understanding complex theoretical computer science concepts through clear explanations, examples, and guided problem-solving
- Supported students in building CRUD applications using JavaScript, HTML, and CSS, helping them apply foundational web development principles in hands-on projects

Projects

Open Source Contributions github.com/dandeandean

- Building the cloud native ecosystem: YokeCD, Terraform Docs, Hashicorp, and Postgres Operator
- Identified and remediated bugs in a Traefik upstream: Oxy
- Bug fixes for LSP crashes in Go-based cli tooling: Zk-Org
- R&D work for drone orchestration software under Penn State's capstone project: RI-SE
- Asynchronous collaboration on planning, communication, and implementation
- Go software engineering focused on bug fixes refactoring, and Developer Experience

Home Lab github.com/dandeandean/nixos-config

- K3s Kubernetes cluster managed via FluxCD (GitOps) across modular, declaratively configured NixOS nodes
- Self hosted LLM model server for personal development
- Self hosted GitHub Actions Runners on Kubernetes
- Cloudflare Tunnels for DNS configuration
- Cloud-native principles with continuous delivery

Skills

Technology: Azure, Kubernetes (Azure Kubernetes Service, Operators/Controllers), Datadog, Docker, Ansible, CI/CD Pipelines, Linux (Ubuntu, Debian, NixOS), SQL (MySQL, PostgreSQL, Azure), Redis, Terraform/OpenTofu, Packer, Helm, Networking, Traefik, Nginx
Languages: Go, Python, Bash, HCL, Powershell, Nix
Spoken Languages: English (Native), Spanish (Limited Working Capacity), French (Basic)

Education

The Pennsylvania State University August 2023 — May 2024
M.S. Cybersecurity Analytics and Operations *State College, PA*
The Pennsylvania State University August 2020 — August 2023
B.S. Computer Science - Minors Mathematics and French *State College, PA*